

2026-27 全新推出

小五人工智慧與 STEAM 大師課程

# AI 創客勇士

小學五年級（約 10 歲）· 20 堂課 · 網頁應用程式 + AI 編程夥伴

賦能學生探索智慧城市工程，透過互動遊戲訓練自訂機器學習模型，並組裝 2D AI 勇士代理人來治理城市分區。讓學生明白 AI 不是魔法，而是需要被建構、訓練、微調與審計的工具。從第一個數據標籤到最後的成果展覽，全程由學生親自掌控自己的創作。

20

堂課程

17

項 AI 能力

4

大技能領域

1

個可攜式 AI 夥伴

# 課程願景

賦能學生探索智慧城市工程，透過互動遊戲訓練自訂機器學習模型，並組裝 2D AI 勇士代理人來治理城市分區。AI 不是魔法，而是需要被建構、訓練、微調與審計的工具。從第一個數據標籤到最後的成果展覽，全程由學生親自掌控自己的創作。

## 便攜式 AI 編程夥伴

懸浮於 Web App 內的 AI 助手，可索引已解鎖的程式碼區塊並協助自動接線至勇士。支援即時自然語言邏輯組裝、持久記憶追蹤全部技能，一鍵匯出與跨學期匯入。

- 即時自然語言邏輯組裝
- 持久記憶追蹤全部 17 項技能
- 一鍵匯出與跨學期匯入 (.aibid 檔案)
- 由課程提供，已預先設定完成

## 開放式創意沙盒

三層自訂選項讓學生進行原創探索，而非機械式執行指令。

- **市政原型**：環保衛士、急救先鋒或市民管家，各自具備獨特使命
- **萬用提示詞**：自訂規則如「當積水超過 50% 時優先救援寵物」
- **廣播人設**：自訂城市緊急事件時的溝通風格

# 6 步專案流程

每節課程皆遵循一套經過驗證的 6 步節奏，從動手玩遊戲到建立持久的 AI 自主能力。



# 課程範例

每堂課將動手遊玩的 AI 遊戲連結至一項永久的勇士能力。以下為六個實際範例。

| 領域   | 課程       | 勇士能力          | AI 概念         | 整合方式       |
|------|----------|---------------|---------------|------------|
| ● 感知 | L11 觀察感知 | Watch-Sense   | 特徵提取 / 邊緣檢測   | 視覺區塊 → 頭部  |
| ● 動作 | L3 巡航之腿  | Rover-Legs    | 全覆蓋路徑規劃       | 路徑區塊 → 雙腿  |
| ● 動作 | L6 運輸規劃  | Haul-Planner  | 路線最佳化         | 路線區塊 → 背包  |
| ● 規劃 | L13 連鎖預測 | Ripple-Warden | 預測模型 / 二級效應   | 預測區塊 → 核心  |
| ● 規劃 | L15 成本感知 | Cost-Sense    | 模型選擇 / 成本權衡   | 預算區塊 → 核心  |
| ● 倫理 | L12 審判核心 | Judgment-Core | 優先級最佳化 / 倫理決策 | 優先級區塊 → 心臟 |

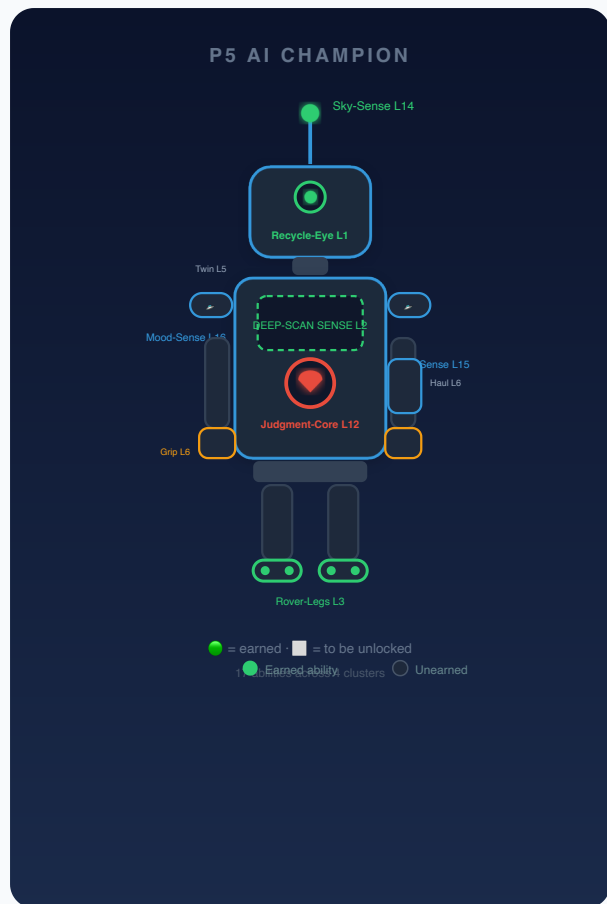
## 從遊戲遊玩到勇士功能整合

從遊玩遊戲到永久升級 AI 勇士的流程分為四個明確步驟，全部在網頁應用程式內完成。



**匯出與重複使用：**學生可隨時點擊「儲存 AI 夥伴」下載勇士檔案（.aibid）。在下學期專案中上傳此檔案即可完整復原全部 17 項功能。

每位學生的勇士是一個視覺化且持續成長的創作。隨著獲得新能力，相應的身體部位會隨之亮起：綠色代表已生效，灰色代表尚未解鎖。



- 頭部： 回收之眼（視覺區塊）· 觀察感知（指標區塊）
- 天線： 天空感知（掃描區塊）
- 胸部： 深層掃描（音訊 / 信號處理區塊）
- 心臟 / 核心： 審判核心（優先級區塊，透過倫理課程獲得）
- 背部： 運輸背包（路線最佳化區塊）
- 肩膀： 雙子無人機·抓取器（機械組件區塊）
- 雙腿： 巡航車輪·巡航之腿（執行器區塊）
- 綠光亮起 = 已獲得能力 · 灰色 = 待解鎖能力

# 全 17 項能力領域矩陣

17 堂課各為勇士增加一項永久能力。

## ● 感知與覺察 (SENSES)

- L1 回收之眼 影像分類

[ 視覺區塊 ]
- L11 觀察感知 檢測閾值

[ 指標區塊 ]
- L14 深層掃描 信號處理

[ 音訊區塊 ]
- L16 情緒感知 情感分析

## ● 動作與執行 (BODY)

- L3 巡航之腿 路徑尋找

[ 執行器 ]
- L4 條件布線 條件邏輯

[ 伺服器 ]
- L5 雙子無人機 多代理人資源分配

[ 機械組件 ]
- L6 運輸規劃 路線最佳化

[ 引擎 ]

## ● 策略與最佳化 (PLANNER)

- L7 交通大腦 網路調度
- L8 成長規劃 預測驅動規劃
- L10 電網衛士 AI 競價監督
- L13 連鎖預測 二級效應
- L15 成本感知 模型選擇與 Token 效率

## ● 倫理與審計 (CONSCIENCE)

- L12 審判核心 優先級最佳化與價值對齊
- L17 審計核心 人機協同審計

總結專案 (L18-L20)：組裝與創意沙盒 → 紅隊測試與審計日誌 → 大師展覽會 Showcase

# 學習成果與評量標準

 **AI 素養與調度能力**

理解數據集平衡、神經網路參數、感測器閾值及自然語言邏輯整合

 **創意主導力與 Prompt 工程**

透過提示 AI 夥伴建立獨特萬用規則，表達原創解題思維

 **倫理審計與價值對齊**

對 AI 代理人進行壓力測試，評估公平偏見、隱私保護與安全覆蓋

 **數位資產所有權與延續性**

體驗匯出並跨學期重複使用個人 AI 夥伴資產

# 總結專案評量標準

| 評量項目      | 權重  | 說明                            |
|-----------|-----|-------------------------------|
| 系統架構與組裝   | 25% | 感知、移動、路徑規劃等程式碼區塊的功能性整合        |
| 創意創新與自訂化  | 20% | 自訂規則、萬用提示詞及市政人設的原創性           |
| 倫理對齊與審計   | 25% | 同儕紅隊測試表現、隱私合規性及安全覆蓋機制         |
| 溝通發表與展覽簡報 | 30% | 3 分鐘簡報清晰度，以及現場問答解釋 AI 機制的表達能力 |

## 勇士發表日盛會

學生於成果發表日展示其完整建構的 AI 勇士。勇士將巡航 AI 城市，逐區展示全部 17 項超能力。每位學生介紹其勇士名稱、設計抉擇、各部位功能及最具成就感的一刻。AI 夥伴在發表前協助演練發言重點，隨後退至幕後。最後可匯出專屬勇士卡片作為可列印的成果總結。

🎓 學年結束時，每位學生都擁有一份屬於自己的創作：一套他們親自訓練、微調、自訂並發表的 AI。

### 小五 AI 創客勇士課程

小學五年級（約 10 歲）· 20 堂課 · 網頁應用程式 + AI 編程夥伴  
屬於 AI 創客勇士課程系列 — 逐課建立 AI 素養

繁體中文

# P5 AI 創客勇士課程



ENGLISH

# P5 AI Champion Programme

★ NEW FOR 2026-27

## P5 AI & STEAM Master Curriculum

# AI Champion

Primary 5 (Age 10) · 20 Lessons · Web App + AI Coding Buddy

Empower students to build, train, and customise their own AI Champion. A humanoid robot with unique abilities spanning perception, movement, planning, and ethical judgment. Each lesson teaches an AI concept through gameplay, then adds a new ability to their Champion. By year-end, every child's AI stands ready to govern a smart city district.

**20**

LESSONS

**17**

AI ABILITIES

**4**

SKILL CLUSTERS

**1**

PORTABLE BUDDY

# Programme Vision

The P5 AI Champion programme empowers students to explore smart city engineering, train custom ML models through interactive games, and assemble a 2D AI Champion Agent to govern urban districts. AI is not magic. It is built, trained, tuned, and audited. Students own their creation from first label to final showcase.

## Portable AI Coding Buddy

A floating AI assistant inside the web app that indexes unlocked code blocks, assists with auto-wiring into the Champion, and can be exported as a downloadable .aibid file to reuse in future projects.

- Real-time natural language logic assembly
- Persistent memory tracking all 17 unlocked skills
- One-click export & re-upload across terms
- Pre-configured by the programme

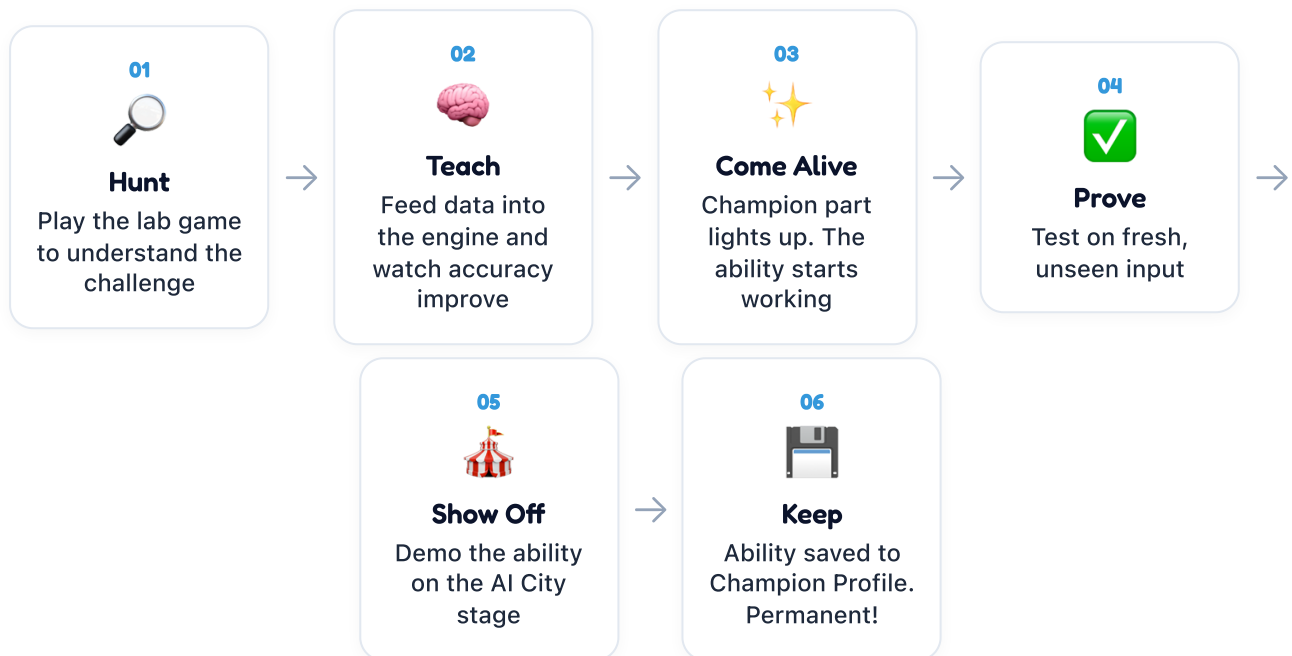
## Open Creative Sandbox

Three customization layers give every Champion a unique identity. Students invent. They don't just follow instructions.

- **Civic Archetypes:** Eco-Guardian, First Responder, or Citizen Concierge, each with unique missions
- **Wildcard Prompts:** "Prioritise pet rescue when flood levels exceed 50%"
- **Broadcast Persona:** Custom communication styles during city emergency events

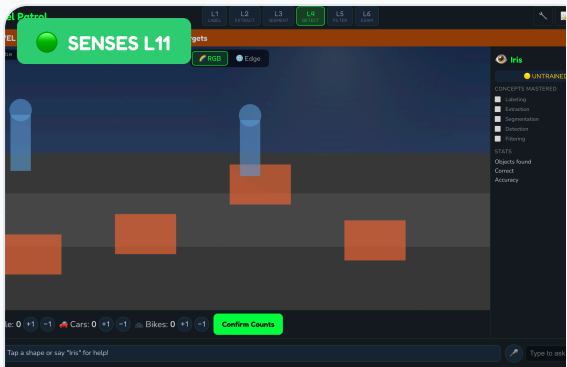
# The 6-Beat Project Flow

Every lesson follows the same proven rhythm. Students learn by doing, from hands-on gameplay to permanent AI ownership.



# How a Lesson Works

Each lesson connects a hands-on AI game to a permanent Champion ability. Here are six real examples: three featured below, three more in the compact grid.



## Watch-Sense

**AI Concept:** Feature Extraction / Edge Detection. Students toggle between a cluttered RGB view and clean Edge Mode to find hidden objects. AI strips away irrelevant detail (shadows, glare, colour) to focus on the shapes that matter.

**Champion Ability:** Watch-Sense lets the Champion detect moving objects in noisy environments, spotting a person in rain or a car in heavy glare.

 Game →

 [Vision Block] earned →

 Wired into Champion Head

### Ripple-Warden

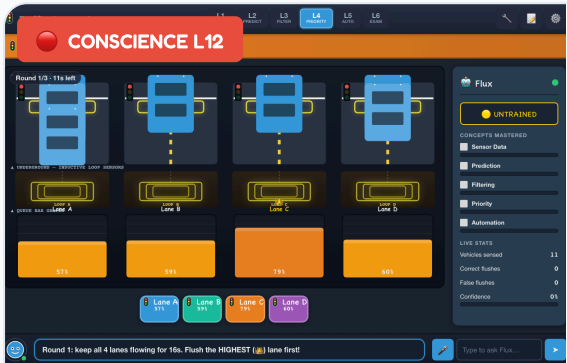
**AI Concept:** Predictive Modeling / Second-order effects. Translucent bands show incoming traffic 5 seconds ahead. Every adjustment to green-light timing ripples through the system, because a switching cooldown means you must commit.

**Champion Ability:** The Ripple-Warden foresees consequences: if I give more time to this lane, how long before the other overflows?

 Game →

 [Predictive Block] earned →

 Wired into Champion Core



## Judgment-Core

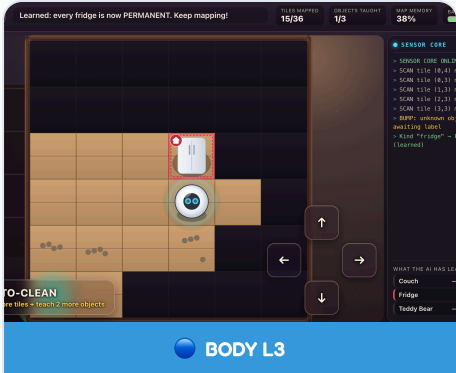
**AI Concept:** Priority Optimisation / Ethical Decision-Making. Four lanes compete for one green light. Flush the wrong lane and resources are wasted. The Flux dashboard shows the ethical stakes of wasted cycles vs. overflows.

**Champion Ability:** Judgment-Core weighs competing demands and makes the fairest call when multiple tasks need attention.

 Game →

 [Priority Block] earned →

 Wired into Champion Heart

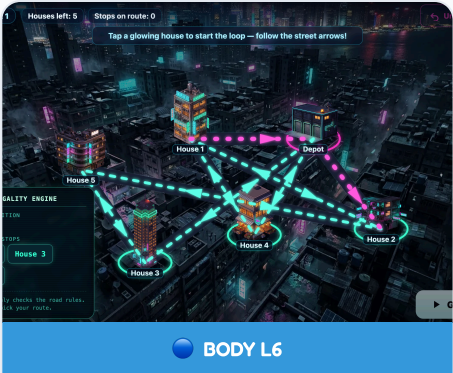


### Rover-Legs

**AI Concept:** Complete-coverage pathfinding. Navigate a room with obstacles, cleaning every tile without repeats.

**Champion:** Rover-Legs uses the same algorithm to navigate the city without collisions.

- Game →
- [Pathfinder Block] →
- Wired into Legs

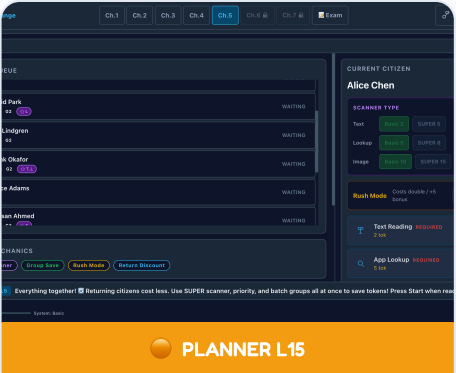


### Haul-Planner

**AI Concept:** Route optimisation under weight and battery limits. Plan the shortest loop that serves all stops.

**Champion:** Haul-Planner optimises delivery and logistics routes so nothing is wasted.

- Game →
- [Route Block] →
- Wired into Backpack



### Cost-Sense

**AI Concept:** Model selection / Cost trade-offs. Choose the right AI engine for each task, balancing quality against token budget.

**Champion:** Cost-Sense chooses the most efficient model for each job within resource limits.

- Game →
- [Budget Block] →
- Wired into Core

# From Game Play to Champion Integration

The flow from playing a game to permanently upgrading your Champion happens in four clear steps, entirely inside the web app.



## 1. Play the Lab Game

Complete the lesson's game by labelling data, tuning sensors, or solving an optimisation challenge.



## 2. Earn a Code Block

The game completion screen shows the earned code block. It appears in the student's inventory.



## 3. Wire Into Champion

Open the Champion Builder. Drag the block from inventory onto the matching body part slot.



## 4. Part Glows – Ability Active

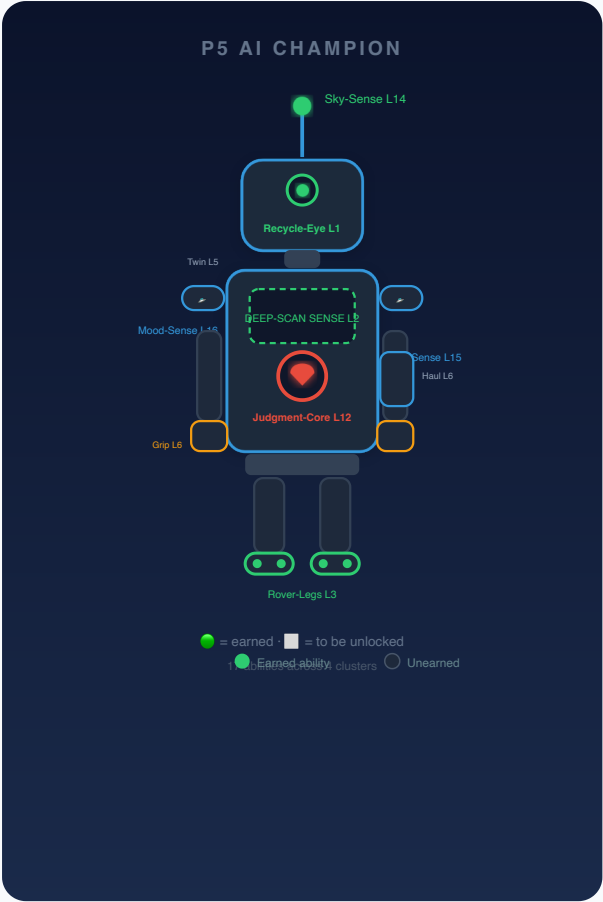
The Champion part lights up green, saved permanently to the Champion Profile.



**Export & Reuse:** At any time, students click "Save AI Buddy" to download their Champion profile (.aibid file). Uploading this file in next term's project instantly restores their companion with all 17 abilities intact.

# The AI Champion

Each student's Champion is a visible, growing creation. As abilities are earned, body parts light up. Green means working, grey means still to unlock.



- Senses:** Head (Recycle-Eye) · Chest (Deep-Scan) · Antenna (Sky-Sense)
- Body:** Legs (Rover Wheels) · Shoulders (Twin Drones) · Arms (Grippers)
- Heart:** Core (Judgment-Core), earned through ethical dilemma lessons
- Back:** Haul Backpack (Route Optimiser)
- Glowing = ability earned · Grey = to be unlocked

# All 17 Abilities by Cluster

Each of the 17 lessons adds one permanent ability to the student's AI Champion.

## SENSES – Perception & Awareness

|     |             |                      |                |     |                 |                   |                 |
|-----|-------------|----------------------|----------------|-----|-----------------|-------------------|-----------------|
| L1  | Recycle-Eye | Image classification | [Vision Block] | L2  | Deep-Scan Sense | Signal processing | [Audio Block]   |
| L11 | Watch-Sense | Detection thresholds | [Metric Block] | L14 | Sky-Sense       | Clustering        | [Scanner Block] |
| L16 | Mood-Sense  | Sentiment scoring    |                |     |                 |                   |                 |

## BODY – Movement & Action

|    |             |                        |             |    |               |                    |          |
|----|-------------|------------------------|-------------|----|---------------|--------------------|----------|
| L3 | Rover-Legs  | Pathfinding            | [Actuator]  | L4 | Reflex-Wiring | Conditional logic  | [Servo]  |
| L5 | Twin-Drones | Multi-agent allocation | [Mechanism] | L6 | Haul-Planner  | Route optimisation | [Engine] |

## PLANNER – Strategy & Optimisation

|     |               |                               |  |     |              |                                   |  |
|-----|---------------|-------------------------------|--|-----|--------------|-----------------------------------|--|
| L7  | Transit-Brain | Network scheduling            |  | L8  | Grow-Planner | Forecast-driven planning          |  |
| L9  | Flow-Planner  | Load-shifting, safety reserve |  | L10 | Grid-Warden  | AI bidding oversight              |  |
| L13 | Ripple-Warden | Second-order effects          |  | L15 | Cost-Sense   | Model selection, token efficiency |  |

## CONSCIENCE – Ethics & Audit

|     |               |                                        |  |     |              |                            |  |
|-----|---------------|----------------------------------------|--|-----|--------------|----------------------------|--|
| L12 | Judgment-Core | Priority optimisation, value alignment |  | L17 | Auditor-Core | Human-in-the-loop auditing |  |
|-----|---------------|----------------------------------------|--|-----|--------------|----------------------------|--|

**Capstone (L18-L20):** Assembly & Creative Sandbox → Red-Teaming & Audit Log → Grand Expo  
Showcase with downloadable .aibid profile

# Learning Outcomes

### AI Literacy & Orchestration

Understand dataset balance, neural net parameters, sensor thresholds, and natural language logic integration.

### Creative Agency & Prompt Engineering

Express original problem-solving ideas by prompting the AI Buddy to build unique wildcard rules.

### Ethical Auditing & Alignment

Stress-test agents for fairness bias, privacy protection, and safety override reliability.

### Digital Ownership & Continuity

Experience exporting and reusing digital AI companions across multi-term projects.

# Capstone Assessment Rubric

| Component                           | Weight | Description                                                                |
|-------------------------------------|--------|----------------------------------------------------------------------------|
| System Architecture & Assembly      | 25%    | Functional integration of perception, locomotion, pathfinding code blocks  |
| Creative Innovation & Customisation | 20%    | Originality of custom rules, wildcard prompts, civic persona design        |
| Ethical Alignment & Audit           | 25%    | Performance under peer red-teaming, privacy compliance, safety overrides   |
| Communication & Expo Pitch          | 30%    | Clarity of 3-minute pitch, ability to explain AI mechanics during live Q&A |

## Champion Reveal Day

Students present their fully-built AI Champion at a showcase event. The Champion tours the AI City, demoing all 17 powers district by district. Each student presents their name, design choices, what each part does, and one proud moment. The AI Buddy helps rehearse talking points beforehand, then steps back. The Champion Card is exported as a printable summary.

🎓 By year-end, every child has built something they own: an AI they trained, tuned, customised, and presented.

### P5 AI Champion Programme

Primary 5 (Age 10) · 20 Lessons · Web App + AI Coding Buddy  
Part of the AI Champion programme — building AI literacy lesson by lesson